Capacitively-Coupled Chopper Instrumentation Amplifier for Linear Hall Effect Sensor

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Team Chop

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Contents

1 Project Information

Project Name	CCIA for Linear Hall Effect Sensor
Team Members	alang
Review Date	
Review Version	
Track	B

2 Design Objective

The aim of this project is to design and characterize a monolithic magnetic sensing system based on the Hall Effect. If tapeout is completed, the specifications of the sensor and readout circuit will be characterized individually.

The target specifications for the magnetic sensing system are shown in Table 1. These are currently guess-timated based on what has been achieved in the literature and are subject to change.

Specification	**Value**
Maximum Offset	$100\,\mu T$
Area	$0.25\,mm^2$
Resolution	$100\,\mu T_{rms}$
Input Range	$\pm 100\,mT$
Bandwidth	10 kHz
Supply Voltage	3.3 V
Maximum Supply Current	2 mA
Temperature Range	$-40\,°C$ to $125\,°C$
Clock Frequency	100 kHz
Output Common-Mode Voltage	1.65 V

Table 1: System Specifications

The motivation for this project in the context of the Chipathon program is to provide reusable IP to the open-source silicon community. Priority will be given to designing a simple system that can be well characterized as opposed to a novel system. Overall, if the tapeout is successful, future designers would be able to reuse and customize both the sensor design and the readout circuit for their specific application. In any case, future Chipathon participants will be able to learn from the successes and failures of the project.

3 System Overview

A block diagram of the system is shown in Fig. 1.

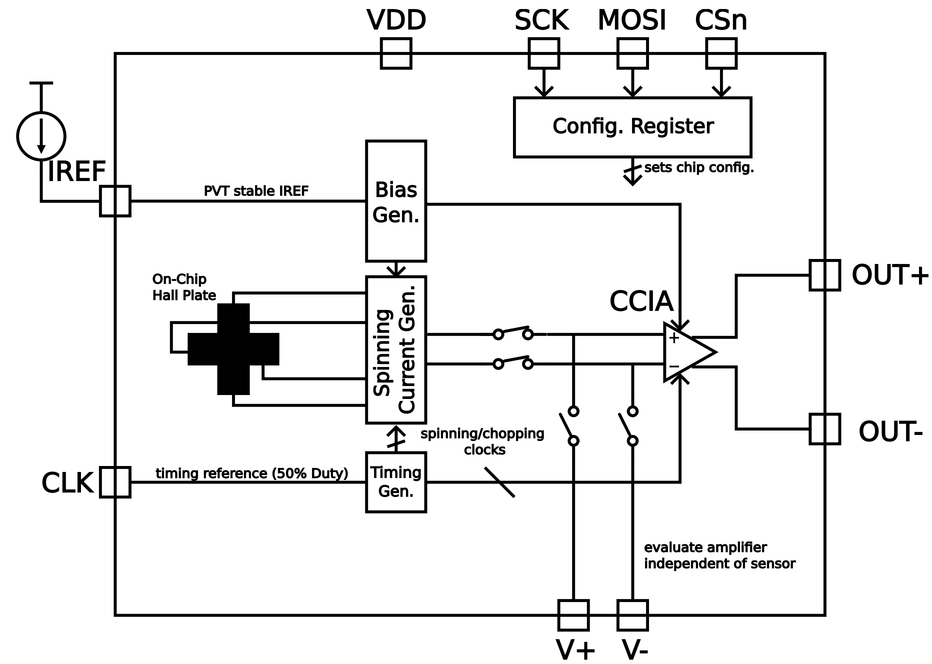

Figure 1: Block diagram of the proposed architecture.

In the diagram, an on-chip horizontal Hall plate converts an incoming magnetic field (perpendicular to plane of chip) into a small voltage that is amplified by the instrumentation amplifier (IA) prior to being driven to the DUT output pins (OUT+/OUT-). The entire design operates from a single supply. Aside from the SPI configuration interface (only used during idle periods), the entire design operates in a single synchronous clock domain derived from the external clock signal (CLK). Two additional pins are used as analog debug pins (V+/V-). Based on the system configuration, these pins will used to probe various points within the design to enable individual characterization of the sensor and IA.

A full pinout table of the design is given in Table 2.

Analog Pins		
IREF	Analog	Current reference
OUT+	Analog	(+) Output
OUT-	Analog	(-) Output
VTP+	Analog	(+) Test Point
VTP-	Analog	(-) Test Point
Digital Pins		
SCK	Digital Input	SPI configuration clock
MOSI	Digital Input	SPI configuration data input
CSn	Digital Input	SPI chip select
CLK	Digital Input	Chopping/spinning clock reference
Supply Pins		
VDD	Power	Supply input
VSS	Power	Ground (shared with other Chipathon blocks)

Table 2: Project I/O description.

4 Schematic Summary

This section describes the schematic-level operation of each part of the design. For each block, both behavioral and transistor-level models are described, including motivation for key design decisions. For all blocks, the transistor-level sizing is subject to change during the verification process.

A more detailed block diagram of the top level schematic is shown in Fig. 2, using the signal names used throughout the Xschem schematic design. The sub-sections below describe the detailed operation of each block.

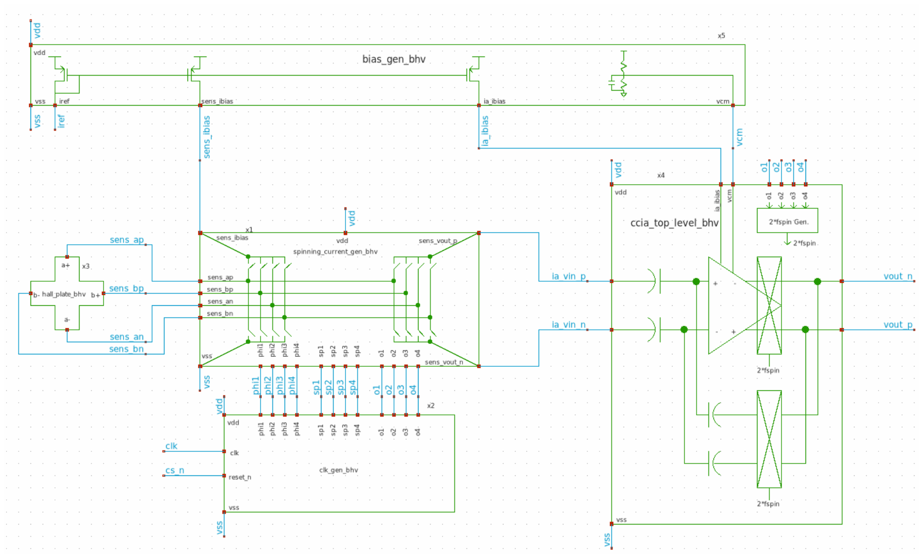

Figure 2: More detailed block diagram of top level schematic, showing the detailed connection between all blocks, excluding the SPI configuration interface. The Hall plate (left) is connected to the spinning current generator (center), which periodically swaps the bias/readout terminals of the sensor prior to amplification and demodulation by the instrumentation amplifier (right). The internal clocks are generated by the clock generator block (bottom) and all bias currents are derived from an external reference in the bias generator block (top).

4.1 On-Chip Hall Plate

Purpose

The on-chip Hall plate is a four-terminal device. Throughout the design, the naming convention given in [1] is used to refer to the four terminal device. The orientation of the four pins, (a+, b+, a-, b-), is shown in Fig. 3. As hinted in the schematic symbol, a cross-shaped plate geometry is used to enable four-phase spinning current operation, described in Section 4.3.

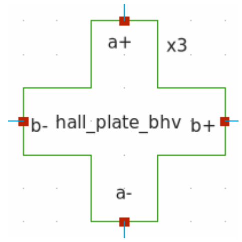

Figure 3: Xschem schematic symbol for on-chip Hall plate, showing (a+, b+, a-, b-) pin orientation and naming convention.

Inputs and Outputs

As discussed in the previous section, the Hall plate has only four terminals since the bias and readout is performed in a higher-level block.

INSERT I/O TABLE

Sub-sheets

- Behavioral: `xschem/hall_plate_bhv/hall_plate_bhv.sch`
- Implementation: ×

Behavioral Model

The plan for sensor behavioral modeling is to use the basic model linear model described in [2]. Clearly, the symbol shown in Fig. 3 needs an additional input to provide the incoming magnetic field.

Design Methodology

- n-well used to maximize sensitivity
- Design equations in [2] used to estimate bridge resistor values in behavioral model from n-well sheet resistance and geometry

Transistor-Level Implementation

The final schematic for Hall-Plate will simply have pins for the four terminals for DRC/LVS purposes. All verification will be performed with the behavioral model so that the effect of an external magnetic field can be simulated.

4.2 Bias Generation

Purpose

The purpose of the bias generation block is to produce the bias currents and voltages needed by each block from an off-chip reference current. A conceptual schematic of the bias generation block is shown in Fig. 4.

Figure 4: Xschem schematic symbol for bias generation block. The block creates the sensor bias current and IA current reference from an off-chip reference current source. Also, the IA common-mode voltage is generated by an on-chip resistor divider.

Inputs and Outputs

INSERT I/O TABLE

Sub-sheets

- Behavioral: `xschem/bias_gen_bhv/bias_gen_bhv.sch`
- Implementation: ×

Behavioral Model

The behavioral model will use ideal current-controlled current sources (CCCS) to generate the required bias currents and ideal passive components to generate the IA common-mode voltage.

Design Methodology

As hinted in the schematic symbol, current mirrors will be used to produce the bias currents. Transistor dimensions will be chosen to minimize short-channel effects and current mismatch.

The common-mode voltage generator will be evaluated initially using only passive components (resistor values chosen to achieve satsifactory tradeoff between silicon area and static power dissipation), but this design decision may be revisited if needed.

Transistor-Level Implementation

Not started: ×

4.3 Spinning Current Generation

Purpose

The purpose of the spinning current generation block is to periodically switch the roles of the four sensor terminals between BIAS+, BIAS-, SENSE+, and SENSE-. The switching sequence is chosen to separate the hall voltage from the sensor offset and 1/f noise (the hall voltage is modulated to high frequency while the offset and 1/f noise remains at DC). This allows the sensor offset and 1/f noise to be largely removed from the hall voltage in later processing stages.

A conceptual schematic diagram of the spinning current generation block is shown in Fig. 5. The naming convention for spinning clocks is stolen from [3], where *sp(x)* clocks show the active phase, *phi(x)* clocks are used to connect the sensor bias currents to the BIAS terminals, and *o(x)* clocks are used to connect the SENSE terminals to the IA input. Separate clocks are used to connect the BIAS and SENSE terminals to avoid adding additional spikes to the IA input from the bias current switching (time is given for the SENSE voltage to settle prior to connection to the IA).

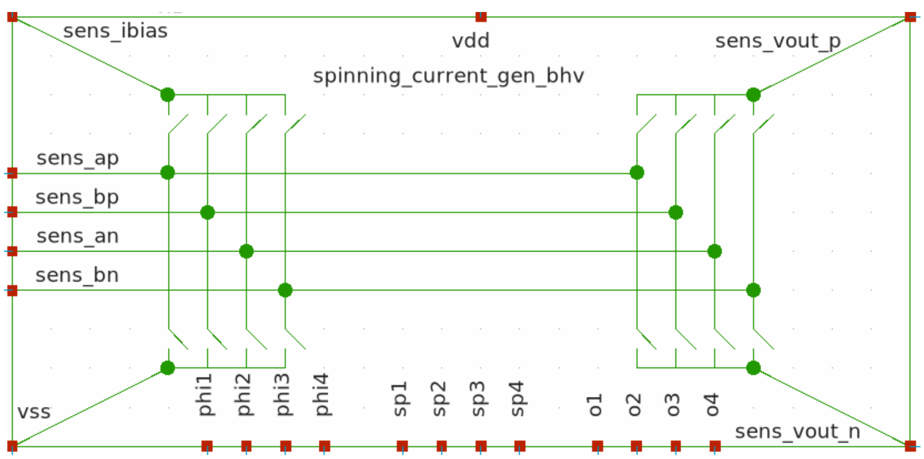

Figure 5: Xschem schematic symbol for spinning current generation block.

Inputs and Outputs

INSERT I/O TABLE

Sub-sheets

- Behavioral: `xschem/spinning_current_gen_bhv/spinning_current_gen_bhv.sch`
- Implementation: ×

Behavioral Model

Ideal switches will be used to implement the eight switches needed to connect the sensor to the IA.

Design Methodology

Transmission gates will be used to implement the SENSE and BIAS switches. Consideration will be given to both on-resistance and charge injection (ADD REF).

Transistor-Level Implementation

Not started: ×

4.4 Timing Signal Generation

Purpose

The purpose of the timing signal generation block is to produce the *sp(x)*, *phi(x)*, and *o(x)* clocks from an external clock. Descriptions of the three clock signals are given in Section 4.3. The external interface of the timing signal generation block is shown in Fig. 6

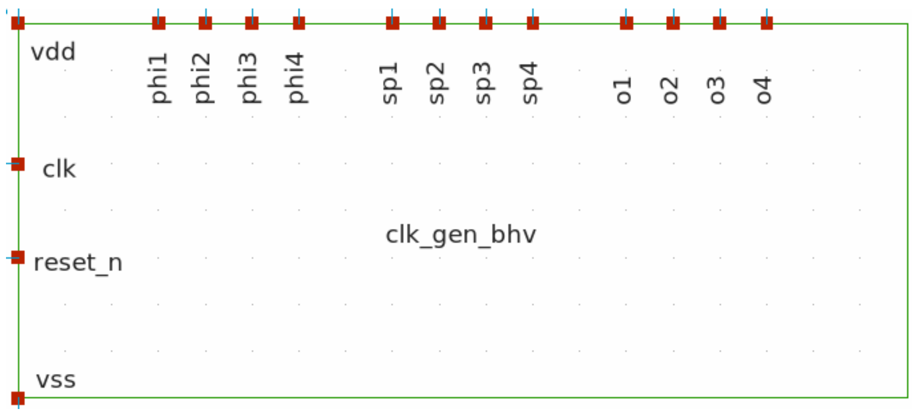

Figure 6: Xschem schematic symbol for timing signal generation block.

Inputs and Outputs

INSERT I/O TABLE

Sub-sheets

- Behavioral: `xschem/clk_gen_bhv/clk_gen_bhv.sch`
- Implementation: ×

Behavioral Model

Standard cells may be used right away to implement the clock generation logic. If this slows down simulation significantly, time will be invested to use the ngspice XSPICE primitives to implement the required digital circuit in an event-driven paradigm.

Design Methodology

The clock generation circuits from Chapter X of [3] will be used.

Transistor-Level Implementation

Not started: $\times$

4.5 Configuration Register

The configuration register will be a N-bit shift register, allowing for the dynamic configuration of the test chip. Some ideas of configuration options are: (1) variable sensor bias current, (2) preamplifier voltage input (i.e., disconnect sensor and supply IA input externally), (3) sensor debug (i.e., connect hall plate terminals to external pins for debug). Once more of the implementation is completed, these configuration options will be revisited.

4.6 Instrumentation Amplifier

The IA will use a CCIA structure very similar to the one reported in [4].

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

5 Design Assumptions

The following assumptions are made within the design.

- The circuit shall be designed to operate from a single 3.3V supply
- The circuit shall be designed to operate over the temperature range of ($-40\,^{\circ}$C - $125\,^{\circ}$C)
- The specifications of each circuit shall be simulated over all worst-case process corners
- Each analog output (both OUT+ and OUT-) of the readout circuit shall be capable of driving a 10 pF off-chip load (representative of commericially available CMOS op amps).
- To test the full system, an external magnetic field will be applied via a Helmholtz coil pair. A printed circuit board will be designed to supply all DUT inputs and condition the differential readout circuit output prior to digitization.

6 Verification Status

6.1 Functional Simulations

Not started

6.2 Corner Simulations

Not started

6.3 Monte Carlo

Not started

6.4 Sign-off

- ERC Status $\times$
- DRC Status $\times$
- LVS Status $\times$

7 Design Checklist

7.1 Functionality

- Requirements implemented ×
- Reset behavior verified ×
- Startup verified ×
- Power sequencing considered ×

7.2 Analog

- Bias currents checked ×
- Device operating regions verified ×
- Stability analyzed ×
- Noise considered ×
- Matching requirements identified ×

7.3 Digital

N/A

7.4 Mixed-Signal

N/A

7.5 Reliability

- Device voltage limits checked ×
- Current density reviewed ×
- Latch-up prevention considered ×
- ESD strategy identified ×

7.6 Documentation

- Nets clearly named ×
- Hierarchy consistent ×
- Parameters documented ×
- Symbols match implementation ×

8 Open Issues

List of open issues with:

- Description
- Impact
- Proposed Solution
- Owner

9 Questions for Reviewers

10 Review Outcome

11 Appendix

References

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[3] J. Jiang, "Cmos wide-bandwidth magnetic sensors for contactless current measurements," Ph.D. dissertation, Delft University of Technology, 2019. [Online]. Available: https://doi.org/10.4233/uuid:38dfa450-82c2-4b63-92d1-26b2eb3adeed

[4] H. Zou, K.-M. Lei, R. P. Martins, and P.-I. Mak, "A cmos hall sensors array with integrated readout circuit resilient to local magnetic interference from current-carrying traces," *IEEE Sensors Journal*, vol. 23, pp. 16 145–16 153, 2023. [Online]. Available: https://ieeexplore.ieee.org/document/10147816